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Sources and Transmission of Country Risk

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1 Big picture

- **Question.** How can we measure country risk directly, identify the source of country-risk spikes, and trace the transmission of these risks across borders?
- **Core innovation.** The paper uses textual analysis of more than 300,000 English-language earnings conference-call transcripts to measure how much risk each firm associates with each country in each quarter.
- **Unit of observation.** The primitive is firm–country–quarter risk: how much firm i discusses risks associated with country c in quarter t .
- **Main measurement advantage.** Because the measure is granular, it can be flexibly aggregated by origin country, destination country, firm location, industry, foreign versus domestic firms, financial versus non-financial firms, and time.
- **Country-risk result.** Perceived country-risk spikes correspond to recognizable events: sovereign debt crises, political shocks, natural disasters, trade disputes, financial crises, and pandemics.
- **Transmission result.** In normal times, country-risk transmission follows a gravity pattern: firms worry more about risk in countries that are closer, linguistically connected, or historically linked.
- **Crisis result.** During crises, transmission can deviate sharply from normal gravity. Some crises become global, some transmit bilaterally to historically exposed countries, and some transmit irregularly to unexpected destinations.
- **Financial-transmission result.** Developed-market and sovereign-debt crises often transmit more through foreign financial firms. Emerging-market and real-side crises often transmit more to traditionally exposed non-financial firms.
- **Capital-flow result.** Elevated perceived country risk is associated with lower capital inflows and a higher probability of sudden stops.
- **Whose perceptions matter?** The perceptions of foreign firms and financial firms are especially informative for capital flows, particularly portfolio flows.
- **Economic takeaway.** Country risk is not just a country-level macro variable. It is a network object: it is perceived by particular firms, originates in particular countries, and transmits through specific international and financial channels.

2 Data and measurement

2.1 Conference-call data

- The core data are 306,589 English-language earnings conference-call transcripts from Refinitiv Eikon over 2002–2020.
- The transcripts cover 12,326 publicly listed firms headquartered in 84 countries.
- The paper focuses country-risk measurement on the 45 largest economies, together covering more than 90% of world GDP.
- A typical call contains a management presentation and an analyst question-and-answer section, which makes it useful for observing both managerial and investor concerns.
- Text is cleaned by removing metadata and non-alphabetic characters, while preserving capitalization to avoid confusing country names such as Turkey with common words.

Interpretation: The identifying content is not the existence of a firm in a country, but the topics that managers and analysts choose to discuss. Conference calls reveal what risks decision makers view as relevant enough to mention publicly.

2.2 Country training libraries

- The paper builds country-specific dictionaries using Economist Intelligence Unit Country Commerce reports.
- Each country library contains bigrams and phrases that are especially diagnostic of that country.
- The method captures direct country names and country-related institutions, cities, agencies, laws, and political references.
- This is more comprehensive than counting only country names because firms often discuss risks using local institutions or proper nouns rather than the country name itself.

Interpretation: The method moves beyond keyword searches. It allows discussions of “the AKP,” “METI,” “JFTC,” or “Syriza” to count as country-specific risk when those phrases are statistically linked to Turkey, Japan, or Greece.

2.3 Firm-country-quarter country risk

Let b index bigrams in firm i ’s conference call in quarter t , B_{it} be the number of bigrams, r be the position of the nearest risk or uncertainty synonym, and $\omega(b, c)$ be the weight connecting bigram b to country c . The main measure is

$$\text{CountryRisk}_{i,c,t} = \frac{1}{B_{it}} \sum_b \{ \mathbf{1}[|b - r| \leq 10] \times \omega(b, c) \}. \quad (1)$$

Interpretation: A country-related phrase contributes to country risk only when it appears close to a synonym for risk or uncertainty. This makes the measure conceptually a second-moment measure rather than a simple exposure or attention measure.

2.4 Firm-country exposure and sentiment

Country exposure is the weighted frequency of country-related language:

$$\text{CountryExposure}_{i,c,t} = \frac{1}{B_{it}} \sum_b \omega(b, c). \quad (2)$$

Country sentiment counts positive and negative tone words around the same country-related bigrams:

$$\text{CountrySentiment}_{i,c,t} = \frac{1}{B_{it}} \sum_b \left(\sum_{g=b-10}^{b+10} S(g) \right) \omega(b, c), \quad (3)$$

where $S(g) = +1$ for positive tone words and $S(g) = -1$ for negative tone words.

Interpretation: Exposure measures whether the country is discussed at all. Sentiment measures first-moment good or bad news. Country risk measures second-moment uncertainty and downside concern. Separating these objects is central to the paper.

2.5 Country-level aggregation

For a set of firms K , aggregate country risk is

$$\text{CountryRisk}_{c,t}^K = \frac{1}{N_K} \sum_{i \in K} \text{CountryRisk}_{i,c,t}. \quad (4)$$

The paper uses different sets K : all firms, foreign firms, domestic firms, financial firms, non-financial firms, U.S. firms, and industry-specific groups.

Interpretation: The same underlying microdata can answer different questions. For example, one can ask whether Greece looks risky to all firms, foreign firms, financial firms, or only firms headquartered in Europe.

2.6 Transmission risk

Transmission from origin country o to destination country d is

$$\text{TransmissionRisk}_{o \rightarrow d,t} = \frac{1}{N_d} \sum_{i \in d} \text{CountryRisk}_{i,o,t}. \quad (5)$$

Because direct firm-level measurement can be noisy, the paper often approximates

$$\text{CountryRisk}_{i,c,t} \approx \text{CountryExposure}_{i,c,t} \times \widehat{\text{CountryRisk}}_{c,t}^{NHQ}. \quad (6)$$

Interpretation: Transmission risk is a destination-country perception measure. It asks: how much do firms located in destination d worry about risks originating in origin o ?

2.7 Global risk

Global risk averages risk perceptions over firms and countries:

$$\text{GlobalRisk}_t = \frac{1}{N_I N_C} \sum_{i \in I} \sum_{c \in C} \text{CountryRisk}_{i,c,t}. \quad (7)$$

Interpretation: This provides a text-based global risk index comparable in spirit to global financial-cycle measures, but built from executives' and investors' own language.

3 Models and empirical specifications

3.1 Validation: country exposure and observed links

The paper validates $\text{CountryExposure}_{i,c}$ by regressing it on indicators for whether firm i is headquartered in, exports to, or has a subsidiary in country c :

$$\text{CountryExposure}_{i,c} = \alpha + \beta_1 \mathbf{1}\{HQ_i = c\} + \beta_2 \mathbf{1}\{Exports_{i,c} = 1\} + \beta_3 \mathbf{1}\{Subsidiary_{i,c} = 1\} + \epsilon_{i,c}. \quad (8)$$

Interpretation: If the NLP measure is meaningful, firms should discuss countries to which they are economically connected. Table 3 shows exactly this.

3.2 Validation: country risk and asset prices

To validate $\text{CountryRisk}_{c,t}$, the paper estimates country-quarter regressions of the form

$$Y_{c,t} = \alpha_c + \tau_t + \beta \Delta \log(\text{CountryRisk}_{c,t}) + \gamma \Delta IHS(\text{CountrySentiment}_{c,t}) + u_{c,t}, \quad (9)$$

where $Y_{c,t}$ is local equity return, change in sovereign CDS spread, or change in realized equity volatility.

Interpretation: Country risk should be associated with falling equity prices, wider sovereign credit spreads, and higher volatility. Sentiment controls make the test more directly about perceived risk rather than good or bad news.

3.3 Crisis identification

A global crisis occurs when GlobalRisk_t exceeds two standard deviations above its sample mean. A local crisis occurs when country c 's country risk exceeds two standard deviations above its country-specific mean, excluding global-crisis quarters.

Interpretation: The threshold is deliberately simple. It gives a transparent way to label events and lets the text snippets identify the source of each spike.

3.4 Crisis transmission regression

For crisis τ in origin country o , the paper compares crisis-period transmission to the normal, non-crisis transmission pattern:

$$\text{TransmissionRisk}_{o \rightarrow d, \tau} = \alpha_{o, \tau} + \beta_{o, \tau} \overline{\text{TransmissionRisk}_{o \rightarrow d, t \notin S^o}} + \varepsilon_{o \rightarrow d, \tau}. \quad (10)$$

- The predicted value at the median normal exposure is called **Global Impact**.
- The slope $\beta_{o, \tau}$ is **Bilateral Transmission**.
- The R^2 is **Regularity of Transmission**.

Interpretation: A high intercept-like global impact means the crisis is broadly felt around the world. A high slope means historically exposed destinations respond disproportionately. A low R^2 means transmission is irregular relative to the usual pattern.

3.5 Financial transmission regression

The paper estimates whether transmission goes more through financial firms:

$$\text{CountryRisk}_{i, o, \tau} - \overline{\text{CountryRisk}_{i, o, t \notin S^o}} = \alpha_{o, \tau} + \alpha_{o, \tau}^{FIN} \mathbf{1}\{Fin(i) = 1\} + \epsilon_{o \rightarrow i, \tau}. \quad (11)$$

Interpretation: If $\alpha_{o, \tau}^{FIN} > 0$, financial firms react more than non-financial firms. This is especially important for sovereign debt and financial crises.

3.6 Capital flows and sudden stops

The capital-flow regressions have the form

$$\text{Flow}_{c, t} = \alpha_c + \tau_t + \beta \text{CountryRisk}_{c, t} + \gamma X_{c, t} + u_{c, t}, \quad (12)$$

where $\text{Flow}_{c, t}$ is total inflows or portfolio inflows. The sudden-stop regressions replace the dependent variable with an indicator for sudden-stop or retrenchment episodes.

Interpretation: These tests ask whether perceived country risk is not only associated with asset prices, but also with cross-border capital allocation and external financing stress.

4 Identification and empirical design

4.1 What makes the measurement credible

- The paper validates country exposure against observed headquarters, exports, and subsidiaries.
- It validates country risk against asset prices, CDS spreads, and volatility.
- The training libraries are built externally from EIU reports rather than hand-crafted around outcomes.
- The text method separates risk from exposure and sentiment.
- The firm-level primitive allows many placebo-like decompositions: domestic versus foreign, financial versus non-financial, and crisis versus non-crisis.

Interpretation: The paper’s core claim is measurement-based rather than causal. It argues that the text measures capture economically meaningful risk perceptions, then uses them to describe transmission and connect perceptions to outcomes.

4.2 What identifies transmission patterns

- Normal-time transmission creates a benchmark exposure pattern.
- Crisis-time transmission is compared against that benchmark.
- The crisis-specific slope, intercept, and R^2 summarize how the crisis deviates from normal risk flows.
- The financial-transmission regression compares financial and non-financial firms facing the same origin crisis.

Interpretation: The design does not require assuming that crises are random. It asks whether the pattern of perceived risk changes in informative ways once the crisis occurs.

4.3 What identifies capital-flow results

- Country fixed effects absorb persistent cross-country differences.
- Time fixed effects or global-risk controls absorb global risk waves.
- GDP growth and sentiment controls separate risk from contemporaneous macro performance and good/bad news.
- Heterogeneous-perception regressions compare risk as perceived by foreign, domestic, financial, and non-financial firms.

Interpretation: The capital-flow tests isolate country-specific risk perceptions. The key finding is that foreign and financial perceptions explain inflows more strongly than domestic perceptions.

4.4 Limitations

- The dataset relies on English-language calls and is therefore strongest for listed firms with English disclosure.
- The method captures risks discussed by global firms, not all risks experienced by households or private firms.
- Small countries with few global exposures can have noisier measures.
- The paper documents associations with capital flows and asset prices; it does not claim randomized causal variation in risk perceptions.

- Text-based measures depend on the training library, transcript coverage, and the assumption that words near risk synonyms proxy perceived risk.

Interpretation: These limitations are also clarifying. The measure is specifically a measure of risk as perceived by globally listed firms and their investors, which is precisely the set of agents likely to influence asset prices and international portfolio flows.

5 Tables and figures

This section keeps the compact format: adjacent tables and figures are placed together when possible. Each exhibit includes a **Read** block and a more detailed **Interpretation** block.

5.1 Table 1 and Figure 1: sample coverage and evolution over time

Read:

- Table 1 reports the number of firms with calls, sales links, world GDP shares, 2019 firm counts, and market-capitalization coverage for the 45 countries.
- The sample has 11,829 firms with earnings calls and covers countries accounting for 91.3% of world GDP in 2019.
- The United States dominates firm coverage with 6,623 firms, but many non-U.S. countries have meaningful market-capitalization coverage.
- Figure 1 shows that coverage rises over time, especially outside the United States.
- By the late 2010s, the data cover a growing number of countries with at least 50% of local market capitalization represented in the call sample.

Interpretation: The table and figure establish external validity. The method does not rely on a narrow U.S.-only sample: coverage is global enough to trace risk perceptions across major economies. The main caution is that English-language transcript coverage is still better for the U.S. and other large English-speaking markets, so the paper must use aggregation and robustness checks carefully.

Economic message: A text-based global risk measure is only useful if the underlying speakers are economically important. Table 1 and Figure 1 show that the speakers in the data represent a large share of listed-company value in many countries.

TABLE 1
Sample selection of firms with earnings calls

	# of firms (all years)	# of sales link (any year)	% of world GDP (2019)	# of firms (2019)	% of 2019 market capitalization
Argentina	20	94	0.68%	16	44.9%
Australia	434	385	1.78%	313	63.6%
Belgium	45	120	0.59%	28	71.5%
Brazil	178	272	2.17%	135	65.8%
Canada	918	886	2.03%	477	90.5%
Chile	31	88	0.31%	24	58.1%
China	349	738	17.08%	165	21.3%
Colombia	16	67	0.38%	15	85.4%
Czech Republic	6	57	0.26%	6	70.5%
Egypt	8	28	0.48%	4	21.8%
France	161	405	3.13%	122	83.4%
Germany	219	698	4.30%	152	84.5%
Greece	41	27	0.24%	19	59.0%
Hong Kong	115	113	0.40%	68	49.7%
Hungary	4	40	0.18%	4	87.9%
India	362	193	3.21%	263	64.6%
Indonesia	18	66	1.25%	9	12.6%
Iran	0	1	0.53%	0	n/a
Ireland	73	90	0.44%	53	79.9%
Israel	114	74	0.42%	58	47.3%
Italy	109	247	2.29%	70	86.9%
Japan	230	595	5.46%	146	39.3%
Malaysia	23	112	0.44%	14	25.6%
Mexico	98	308	1.50%	65	67.9%
Netherlands	104	207	1.00%	67	80.1%
New Zealand	62	85	0.24%	47	74.5%
Nigeria	14	29	0.60%	10	41.7%
Norway	96	102	0.49%	71	83.9%
Pakistan	4	8	0.39%	0	n/a
Philippines	20	61	0.47%	12	19.7%
Poland	32	86	0.68%	26	61.7%
Russia	54	101	1.75%	35	73.1%
Saudi Arabia	3	31	0.81%	2	5.3%
Singapore	56	208	0.42%	35	49.0%
South Africa	96	96	0.43%	76	83.4%
South Korea	45	233	1.96%	31	30.9%
Spain	75	199	1.58%	59	87.2%
Sweden	198	118	0.66%	154	80.2%
Switzerland	125	145	0.91%	97	88.4%
Taiwan	49	179	n/a	26	37.7%
Thailand	24	74	0.55%	19	29.9%
Turkey	27	61	1.19%	24	39.8%

(continued)

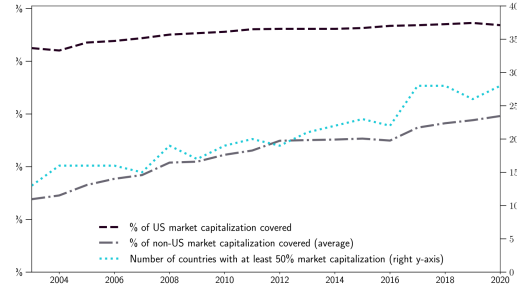


FIGURE 1
Coverage of firms over time

* This figure plots the per cent of market capitalization for firms based in the US and in countries around the world that the sample covers.

(a) Table 1, part 1

(b) Figure 1

TABLE 1
Continued

	# of firms (all years)	# of sales link (any year)	% of world GDP (2019)	# of firms (2019)	% of 2019 market capitalization
U.K.	548	990	3.81%	374	85.0%
U.S.	6,623	1,319	23.81%	3,219	94.6%
Venezuela	2	36	n/a	0	n/a
Total	11,829	10,072	91.3%	6,610	57.5% (mean)

Notes: This table shows for the forty-five countries for which we have text-based measures of country exposure, risk, and sentiment, the number of firms in our data (Column 2), the number of firms that report part of their sales to the country (Column 3), the share in 2019 world GDP (Column 4), the number of firms in our data in 2019 (Column 5), and the percentage of all Compustat's firms' market capitalization by firms in our sample (Column 6). The last row shows the sum for Columns 2–5, and the mean for Column 6. Firms in our sample are all firms for which we have earnings calls between 2002 and 2019; the sales link is taken from Worldscope and defined as the number of firms that report part of their sales to the country at any point between 2002 and 2017; GDP is the real GDP (constant 2015, USD) from the

(c) Table 1, continued

Sample coverage and coverage growth.

5.2 Table 2 and Table 3: text classification and exposure validation

Read:

- Table 2 lists country-specific ngrams for Turkey, Japan, and Greece. Many are institutional, political, legal, or city-related terms rather than simple country names.
- The country weights $\omega(b, c)$ are high for terms that strongly identify a country in the training corpus.

- Table 3 validates country exposure against observed firm-country links.
- Headquarters, export links, and subsidiaries are all strongly related to the text-based exposure measure.
- The coefficient on headquarters is especially large: firms discuss their headquarters country much more than other countries.

Interpretation: Table 2 shows why the algorithm can identify country discussions even when speakers do not use country names. Table 3 shows the measure is not merely linguistic noise: it maps closely to real economic links. This validation is important because all subsequent risk and transmission measures build on the ability to detect country-relevant language.

Economic message: The tables turn text into economic data. Country-linked phrases proxy for commercial exposure; risk language near those phrases turns exposure into perceived risk.

TABLE 2
Top 20 ngrams in the training library of Turkey, Japan, and Greece

Ngram	$\omega(b, c)$	Frequency	Ngram	$\omega(b, c)$	Frequency
PANEL A: TURKEY					
Turkey/Turkish	805.22	2,738	the Undersecretariat	87.61	112
Gazette No	246.57	398	Izmir	82.21	87
Turk Eximbank	171.04	181	the Directive	76.56	135
Ankara	144.58	153	in prioritydevelopment	76.54	81
Official Gazette	131.89	495	prioritydevelopment regions	74.65	79
of Turkey	128.48	187	in Turkey	73.71	78
Istanbul	127.94	244	Region VI	71.18	91
the lira	114.34	121	Undersecretariat of	71.18	91
the GDFI	94.50	100	Patent Institute	70.01	113
an AS	88.63	129	the AKP	68.04	72
PANEL B: JAPAN					
Japan	244.15	7,076	Standards Law	83.63	206
Economy Trade	215.39	466	Japanese	81.28	3,801
the JFTC	207.15	371	Tokyo	81.13	626
Health Labour	138.47	248	Antimonopoly Law	78.70	215
Industry METI	136.24	244	Labour Standards	75.78	207
the METI	115.58	207	AntiMonopoly Law	73.89	182
The JFTC	107.21	192	inhabitant tax	73.49	159
the JPO	86.55	155	Okinawa	72.03	129
the Diet	85.99	154	and Welfare	70.96	246
enterprise tax	84.58	183	Osaka	69.42	171
PANEL C: GREECE					
Greece/Greek	607.83	2,897	The ND	73.09	114
Athens	339.67	640	New Democracy	69.89	109
Hellenic	249.73	649	Greeks	64.75	101
ND government	130.15	203	Strategic Reference	61.55	96
Piraeus	127.91	241	gov gr	61.55	96
Share sale	88.48	138	Attica	59.63	93
an AE	80.78	126	ministerial decisions	59.20	127
Thessaloniki	80.67	152	Alpha Bank	58.34	91
by Law	79.83	511	objective value	57.70	90
the EA	76.30	119	of Development	54.90	236

Notes: This table lists the top twenty ngrams when sorted on $\omega(b, c)$ (the $tf \times idf$ in the training library) for three selected countries. Column 2 shows the $\omega(b, c)$ of the ngram, which is the frequency of the ngram in its country-specific library divided by the total number of ngrams in that library (tf) multiplied by the log of the number of country libraries divided by the number of country libraries that contain the ngram (idf); and Column 3 shows the frequency of the ngram in the country-specific library. A country-specific training library consists of (a) all adjacent two-word combinations (bigrams)

TABLE 3
Country exposure and observed firm links

	CountryExposure _{i,c} (std.)				
	(1)	(2)	(3)	(4)	(5)
1(Headquarter) _{i,c}	2.337*** (0.040)			2.229*** (0.078)	2.943*** (0.102)
1(Exports) _{i,c}		1.225*** (0.025)		1.026*** (0.025)	1.195*** (0.029)
1(Subsidiary) _{i,c}			0.595*** (0.008)	0.262*** (0.006)	0.299*** (0.007)
R ²	0.114	0.059	0.059	0.168	0.205
N	533,925	215,325	387,225	168,570	168,570
Country FE	no	no	no	no	yes

Notes: This table shows coefficient estimates and standard errors from regressions at the firm-country level. All variables are as defined in Section 1; summary statistics are provided in Panel A of Table 4. Column 5 includes country-fixed effects. Standard errors are robust. ***, **, and * denote statistical significance at the 1, 5, and 10% levels, respectively.

(a) Table 2

(b) Table 3

Country-identifying text and validation of exposure.

5.3 Table 4 and Table 5: summary statistics and asset-price validation

Read:

- Table 4 summarizes firm-country exposure, country-risk aggregates, sentiment, firm risk, realized volatility, equity returns, capital inflows, CDS spreads, GDP growth, sudden stops, and the World Uncertainty Index.
- Foreign-firm perceived country risk has a higher mean than domestic perceived country risk, consistent with foreign firms discussing a country only when it is salient.
- Table 5 shows that increases in country risk are associated with negative MSCI equity returns, wider sovereign CDS spreads, and higher realized volatility.
- Controlling for country sentiment does not eliminate the risk coefficients.

Interpretation: Table 4 tells us the panel is rich enough to connect text-based risk to asset prices and capital flows. Table 5 is a crucial validation table: perceived country risk behaves like risk should behave in financial markets. The negative return coefficient says risk spikes coincide with price declines; the CDS coefficient says sovereign credit markets demand compensation; the volatility coefficient says realized uncertainty rises as managers and investors talk more about risk.

Economic message: This is the bridge from language to finance. The text measure is not just attention; it is associated with market repricing of risk.

TABLE 4
Summary statistics

	Mean	Median	St. Dev.	Min	Max	N
PANEL A: FIRM-COUNTRY						
$CountryExposure_{i,c} (std.)$	0.77	0.62	1.00	0.00	82.33	533,925
$1(Headquarter)_{i,c}$	0.02	0.00	0.14	0.00	1.00	533,925
$1(Exports)_{i,c}$	0.06	0.00	0.24	0.00	1.00	215,325
$1(Subsidiaries)_{i,c}$	0.16	0.00	0.36	0.00	1.00	387,225
PANEL B: COUNTRY-QUARTER						
$CountryRisk_{c,t}^{ALL} (std.)$	3.69	3.50	1.00	2.15	10.11	3,240
$CountryRisk_{c,t}^{NHQ} (std.)$	4.22	4.04	1.00	2.57	11.84	3,240
$CountryRisk_{c,t}^{FIN} (std.)$	3.87	3.70	1.00	2.16	11.72	3,240
$CountryRisk_{c,t}^{NFC} (std.)$	3.33	3.12	1.00	1.93	9.89	3,240
$CountryRisk_{c,t}^{US} (std.)$	3.16	2.98	1.00	1.93	10.06	3,240
$CountryRisk_{c,t}^{HQ} (std.)$	0.57	0.24	1.00	0.00	12.27	2,838
$CountrySentiment_{c,t}^{ALL} (std.)$	3.00	2.90	1.00	-0.46	7.40	3,240
$FirmRisk_{i,c,t} (std.)$	3.17	3.00	1.00	0.62	12.25	2,256
$Realized\ MSCI\ volatility_{c,t}$	0.10	0.09	0.06	0.02	1.16	2,961
$MSCI\ equity\ return_{c,t}$	0.02	0.03	0.10	-0.86	0.62	2,958
$Total\ inflows_{c,t} (\%)$	1.66	1.49	2.10	-17.67	20.33	2,936
$Sovereign\ CDS\ spread_{c,t} (pct)$	1.87	0.74	3.92	0.01	29.01	2,713
$Real\ GDP\ growth_{c,t}$	0.93	1.05	5.89	-26.48	29.24	2,882
$1(Stop\ episode\ for\ total\ flows_{c,t})$	0.13	0.00	0.34	0.00	1.00	2,734
$1(Retrenchment\ episode\ for\ total\ flows_{c,t})$	0.13	0.00	0.33	0.00	1.00	2,734
$WUI_{c,t} (std.)$	1.00	0.76	1.02	0.00	9.87	3,240

Notes: This table shows the mean, median, standard deviation, minimum, maximum, and number of observations of all variables that are used in the subsequent regression analyses. Panels A and B show the relevant statistics for the regression sample at the firm-country and country-quarter unit of analysis, respectively. In Panel A, $CountryExposure_{i,c} (std.)$ is the average over time of firm i 's Country Exposure to country c , normalized by the standard deviation; and $1(Headquarter)_{i,c}$, $1(Exports)_{i,c}$, $1(Subsidiaries)_{i,c}$ are binary variables equal to one if firm i is headquartered in country c , reports sales at any point between 2002-2017 to country c , or has at least one subsidiary in country c , respectively. In Panel B, $CountryRisk_{c,t}^{ALL} (std.)$ is the average for country c and quarter t of the Country Risk perceived by all firms as measured in their earnings call transcripts, normalized by the standard deviation in the panel; $CountryRisk_{c,t}^{NHQ} (std.)$, $CountryRisk_{c,t}^{FIN} (std.)$, $CountryRisk_{c,t}^{NFC} (std.)$, $CountryRisk_{c,t}^{US} (std.)$, and $CountryRisk_{c,t}^{HQ} (std.)$ are the same but based on firms not headquartered in c at t , financial (SIC $\in [6000, 6800]$), non-financial (SIC $\notin [6000, 6800]$), U.S.-based, and domestic firms, respectively; $CountrySentiment_{c,t}^{ALL} (std.)$ is the average for country c and quarter t of Country Sentiment perceived by all firms, normalized by the standard deviation in the panel; $FirmRisk_{i,c,t} (std.)$ is the average over all firms headquartered in country c and quarter t of risk words per word mentioned by the firm during its earnings call (restricted to countries for which we have at least five firms), normalized by the standard deviation in the panel; $Realized\ MSCI\ volatility_{c,t}$ is the standard deviation of the daily MSCI stock return for country c during quarter t (based on local currency); $MSCI\ equity\ return_{c,t}$ is the $t-1$ to t change in log of the quarter-average MSCI stock return index (based on local currency) for country c and quarter t ; $Total\ inflows_{c,t} (\%)$ are inflows of equity and debt to country c during quarter t relative to the country's stock of capital in the previous quarter; $Sovereign\ CDS\ spread_{c,t}$ is the end-of-quarter 5-year sovereign CDS spread of country c and quarter t (in per cent); $Real\ GDP\ growth_{c,t}$ is the quarter-to-quarter per cent change in real GDP of country c and quarter t ; $1(Stop\ episode\ for\ total\ flows_{c,t})$ and

(a) Table 4

TABLE 5

Country risk, country sentiment, and asset prices

	$MSCI\ equity\ return_{c,t}$		$\Delta CDS\ spread_{c,t}$		$\Delta Realized\ volatility_{c,t}$	
	(1)	(2)	(3)	(4)	(5)	(6)
$\Delta \log(CountryRisk_{c,t}^{ALL} (std.))$	-0.399*** (0.045)	-0.213*** (0.035)	3.926*** (1.014)	2.768*** (0.805)	0.098*** (0.018)	0.103*** (0.023)
$\Delta HSI(CountrySentiment_{c,t}^{ALL} (std.))$		0.267*** (0.050)		-1.456*** (0.515)		0.008 (0.011)
R^2	0.099	0.230	0.057	0.081	0.015	0.016
N	2,918	2,918	2,626	2,626	2,917	2,917

Notes: This table shows coefficient estimates and standard errors from regressions at the country-quarter level. $HSI(\cdot)$ denotes the inverse hyperbolic sine transformation. All variables are as defined in Table 4; their construction is detailed in Appendix Table 8. Standard errors are clustered at the country level. ***, **, and * denote statistical significance at the 1, 5, and 10% levels, respectively.

(b) Table 5

Summary statistics and validation with asset prices.

5.4 Figures 2–5: sources of country risk

Read:

- Figure 2 decomposes Greek country risk into financial and non-financial perceptions. The major spikes line up with the Greek sovereign debt crisis, bailouts, and the Grexit referendum.
- Figure 3 shows Thailand. The flood disaster and military coup show stronger real-side and non-financial patterns than the Greek sovereign-debt episodes.
- Figure 4 shows U.S. risk perceived by all, domestic, and foreign firms. The Iraq War and Trump election are more salient to foreign firms, while the Fiscal Cliff is more domestic.

- Figure 5 shows global risk. The GFC and coronavirus pandemic exceed the global-crisis threshold, while the European sovereign-debt crisis is visible but below the threshold.

Interpretation: These figures demonstrate why the micro-to-macro design is useful. The same country-risk spike can be decomposed by who perceives it and what text snippets identify its source. Sovereign debt crises look financial. Natural disasters and coups can look more real-side. U.S. risks sometimes matter more to foreigners than to domestic firms, revealing asymmetry in global risk perceptions.

Economic message: Country risk is not homogeneous. Two crises with similar aggregate intensity can transmit through very different channels depending on whether financial firms, domestic firms, or foreign firms are the ones talking.

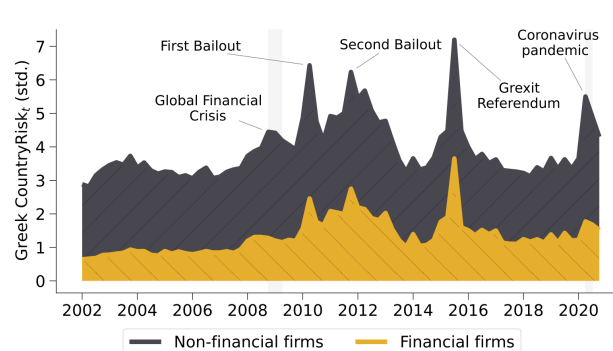


FIGURE 2
Sources of Greek country risk

Notes: This figure plots the time series of Greek $CountryRisk_{i,t}$ as defined in equation (5) but decomposed into Country Risk as perceived by non-financial and financial firms, respectively. The latter are firms whose four-digit SIC code is in 6000–6800. The text excerpts selected from the highest-ranking snippets among all snippets from the top thirty highest-ranked firms when sorted on Country Risk

(a) Figure 2: Greece

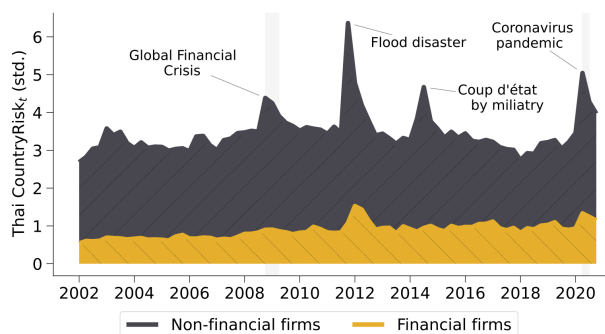


FIGURE 3
Sources of Thai country risk

Notes: This figure plots the time series of Thai $CountryRisk_{i,t}$ as defined in equation (5) but decomposed into Country Risk as perceived by non-financial and financial firms, respectively. The latter are firms whose four-digit SIC code is in 6000–6800. The text excerpts selected from the highest-ranking snippets among all snippets from the top thirty highest-ranked firms when sorted on Country Risk

(b) Figure 3: Thailand

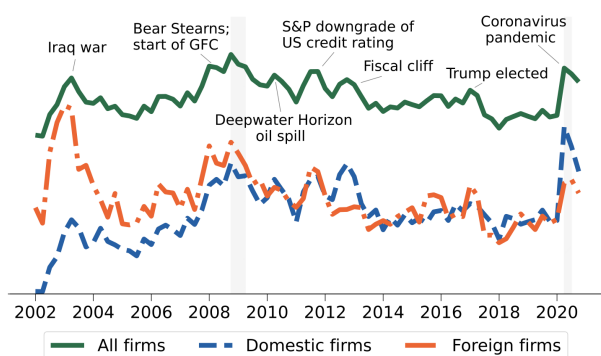


FIGURE 4
Sources and perceptions of U.S. country risk

(c) Figure 4: United States



FIGURE 5
Time series of $GlobalRisk_t$

figure shows the time series of $GlobalRisk_t$ defined as the mean of $CountryRisk_{i,t}$. Marked in grey are the quartile 1 deviations (the red horizontal dashed line), which we define as global crises. The coefficients are standardized to have standard deviation one for 2002q1–2019q4. NBER-based recession quarters are shaded in grey

(d) Figure 5: global risk

Examples of country-risk and global-risk time series.

5.5 Figure 6: local crises and their source labels

Read:

- Figure 6 lists countries with local crises and labels the main event source: Chinese hard landing concerns, Greek sovereign-debt crises, U.S. GFC and Deepwater Horizon risks, Brexit, Thai floods, Fukushima, Hong Kong protests, and others.
- Red dots mark local crises; grey dots mark global crises.

- The figure also indicates whether a crisis transmits disproportionately through financial firms (FIN), non-financial firms (NFC), or irregular channels (I).
- Emerging-market and political/natural-disaster crises often look different from developed-market financial crises.

Interpretation: Figure 6 is a visual crisis atlas. It shows that text-based country risk can systematically recover well-known events without being told what to look for. It also shows the limits of a single global risk factor: many crises are local and have country-specific sources.

Economic message: Policymakers often ask whether a shock is local, global, financial, or real. Figure 6 provides a way to classify that question using the language of firms around the world.

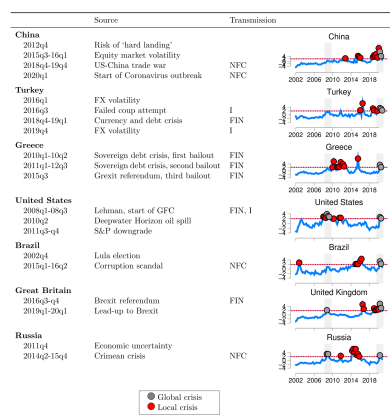


FIGURE 6
For caption see next page.

(a) Part 1

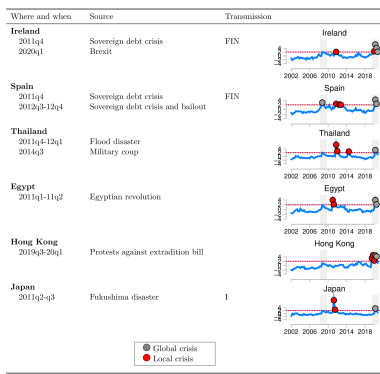


FIGURE 6
For caption see next page.

(b) Part 2

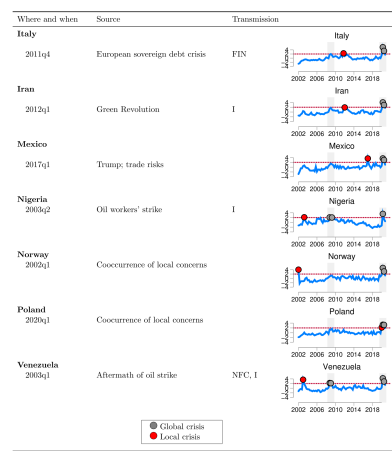


FIGURE 6
Country risk, crises, and patterns of transmission

Note: This table describes and plots country crises based on $CountryRisk_{i,t}$ for the country indicated in Column 1. A global crisis (grey dots in the figures) is defined as $GlobalRisk_{i,t}$ being above two standard deviations over also Figure 5); a local crisis (red dots in the figures) is defined as the country's $CountryRisk_{i,t}$ being above two standard deviations in the past (the red horizontal dashed lines). Column 1 indicates the country and crisis. For Brazil, we assume that 2015q1, which is just below the threshold of two standard deviations, is nevertheless part of the crisis that started in 2015q1. Column 2 indicates the Source of crisis. It is a description summarizing discussions of the crisis above, below, and/or in-between the crisis and the country risk. Column 3 indicates the Transmission of crisis.

(c) Part 3

Figure 6: country risk, crises, and patterns of transmission.

5.6 Table 6 and Table 7: normal transmission and crisis transmission

Read:

- Table 6 lists top origins and destinations of transmission risk. U.S. firms discuss risks from China, Canada, Mexico, Japan, and Brazil; Chinese firms discuss Hong Kong, the U.S., Japan, Taiwan, and Singapore.
- The normal pattern looks like a gravity network: geographic, trade, financial, and historical links matter.
- Table 7 summarizes each crisis using four objects: global impact, bilateral transmission, regularity of transmission, and financial transmission.
- The China coronavirus outbreak has the highest global impact, while the Thai floods and Venezuela oil-strike aftermath have very high bilateral transmission.
- Greek bailout episodes show strongly financial transmission, while some China and Brazil episodes are more non-financial.

Interpretation: Table 6 gives the normal map of perceived exposure. Table 7 shows how crises bend that map. A high global-impact crisis moves many destinations, including those with limited normal exposure. A high bilateral-transmission crisis hits destinations already exposed to the origin. A low regularity crisis

transmits to unexpected places. A high financial-transmission crisis primarily raises risk among financial firms.

Economic message: The crisis transmission framework converts vague terms like “contagion” into measurable components: broad global impact, bilateral amplification, irregularity, and financial-sector concentration.

TABLE 6
Top five origins and destinations of transmission risk for selected countries

Firms headquartered in	Discuss risks from	Risks originating in	Transmit most to
U.S.	China Canada Mexico Japan Brazil Hong Kong	China	Hong Kong Singapore Taiwan South Korea Japan Belgium
China	U.S. Japan Taiwan Singapore China Thailand U.S. Indonesia Singapore	Greece	Italy Spain France Switzerland Turkey Italy Netherlands
Japan	U.S. Russia U.S. Poland Ireland China U.S. Australia Spain	Russia	Chile Spain Mexico France Norway Greece Italy Russia Netherlands
Germany	China Russia U.S. Poland Ireland China U.S. Australia Spain	Brazil	Ireland Australia France Sweden Spain Chile Spain Mexico Brazil Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile
U.K.	U.S. Australia Spain	Turkey	France Sweden Spain Chile Spain Mexico Brazil Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile
India	U.S. Brazil South Africa China Brazil Spain Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile	U.K.	Ireland Australia France Sweden Spain Chile Spain Mexico Brazil Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile
France	U.S. Brazil South Africa China Brazil Spain Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile	Argentina	France Sweden Spain Chile Spain Mexico Brazil Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile
Italy	U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile	Egypt	France Sweden Spain Chile Spain Mexico Brazil Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile
Brazil	U.S. Brazil South Africa China Brazil Spain Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile	Iran	France Sweden Spain Chile Spain Mexico Brazil Italy U.S. Spain Brazil Russia Turkey China Argentina China Colombia Mexico Chile

(continued)

(a) Table 6, part 1

TABLE 7
Crisis transmission patterns

	GLOBAL IMPACT $\hat{\gamma}$	BILATERAL TRANSMISSION $\hat{\beta}_{o,t}$	REGULARITY OF TRANSMISSION R^2	FINANCIAL TRANSMISSION $\hat{\alpha}_{o,t}^{FIN}/\hat{\alpha}_{o,t}$
China: Start of Coronavirus outbreak (2020q1)	3.69	2.58***	0.905	-0.89***
U.S.: Lehman; start of GFC (2008q1-08q3)	2.27	0.92	0.554	2.70**
Japan: Fukushima disaster (2011q2-11q3)	2.12	1.91*	0.281	-0.06
China: U.S.-China trade war (2018q4-19q4)	2.08	1.73***	0.924	-1.03***
China: Equity market volatility (2015q3-16q1)	1.84	1.91***	0.938	0.06
U.S.: S&P downgrade (2011q3-11q4)	1.75	1.01	0.762	0.39
U.S.: Deepwater Horizon oil spill (2010q2)	1.63	0.93	0.673	-0.98
China: Risk of “hard landing” (2012q4)	1.55	1.41***	0.964	-0.68
Greece: Grexit referendum (2015q3)	1.50	2.82***	0.712	4.90***
Mexico: Trump; trade risks (2017q1)	1.44	1.45***	0.793	-0.48
Thailand: Flood disaster (2011q4-12q1)	1.40	4.00***	0.683	0.23
Turkey: Failed coup attempt (2016q3)	1.39	1.44*	0.467	0.33
U.K.: Brexit referendum (2016q3-16q4)	1.38	1.51***	0.857	1.44***
Russia: Crimean crisis (2014q2-15q4)	1.35	2.68***	0.881	-0.47*
Brazil: Corruption scandal (2015q1-16q2)	1.29	1.68***	0.915	-0.89**
Venezuela: Aftermath of oil strike (2003q1)	1.18	5.09*	0.304	-0.88
Greece: First bailout (2010q1-10q2)	1.17	2.80***	0.734	2.83***
Turkey: Currency and debt crisis (2018q4-19q1)	1.16	1.79***	0.628	2.10*
U.K.: Lead-up to Brexit (2019q1-20q1)	1.14	1.17**	0.855	-0.22
Thailand: Military coup (2014q3)	1.02	1.79***	0.856	-1.04
Nigeria: Oil workers’ strike (2003q2)	1.02	1.95	0.380	-0.95
Russia: Economic uncertainty (2011q4)	1.01	1.42***	0.822	-0.51
Greece: Second bailout (2011q1-12q3)	1.00	3.13***	0.722	7.73***
Turkey: FX volatility (2019q4)	0.98	0.97	0.502	-1.22

(b) Table 7, part 1

TABLE 6
Continued

Firms headquartered in	Discuss risks from	Risks originating in	Transmit most to
Canada	U.S. China Mexico Australia U.K.	Japan	South Korea Hong Kong Israel Singapore Switzerland

Notes: This table lists the ten largest economies in which firms in our sample are headquartered (Column 1), the top five countries those firms discuss risks about (Column 2); it also lists ten selected countries that firms perceive risk in about (Column 3), the top five countries those firms are headquartered in (Column 4). For additional countries, see Appendix Table 3. The rankings in Columns 2 and 4 are based on an appropriate sorting of $TransmissionRisk_{o \rightarrow d,t}$ by o for a given d (Column 2) or by d for a given o (Column 4). In Column 4, we exclude countries in which we have fewer than twenty-five firms headquartered, or countries that are not in our list of forty-five countries for which we have measures of $CountryRisk_{o,t}$.

(c) Table 6, continued

TABLE 7
Continued

	GLOBAL IMPACT $\hat{\gamma}$	BILATERAL TRANSMISSION $\hat{\beta}_{o,t}$	REGULARITY OF TRANSMISSION R^2	FINANCIAL TRANSMISSION $\hat{\alpha}_{o,t}^{FIN}/\hat{\alpha}_{o,t}$
Spain: Sovereign debt crisis (2011q4)	0.97	1.55***	0.906	2.24**
Ireland: Brexit (2020q1)	0.97	0.98	0.751	-0.15
Spain: Bailout (2012q3-12q4)	0.97	1.67***	0.884	1.14
Turkey: FX volatility (2016q1)	0.96	1.08	0.603	-0.28
Egypt: Egyptian revolution (2011q1-11q2)	0.92	3.49***	0.902	0.11
Ireland: Sovereign debt crisis (2011q4)	0.90	1.10	0.874	11.85***
Hong Kong: Protests against extradition bill (2019q3-19q4)	0.86	1.49***	0.938	0.79
Italy: Sovereign debt crisis (2011q4)	0.81	1.26***	0.894	3.32***
Iran: Green Revolution (2012q1)	0.80	1.21	0.579	-0.96

Notes: This table lists four characteristics of each local crisis defined in Figure 6: Global Impact, Bilateral Transmission, Regularity of Transmission, and Financial Transmission. The first three characteristics are based on a regression of $TransmissionRisk_{o \rightarrow d,t}$ on $TransmissionRisk_{o \rightarrow d,t,FS}$ as defined in equation (8). Global Impact is the predicted value of $TransmissionRisk_{o \rightarrow d,t}$ for the country with the median of Transmission Risk, $TransmissionRisk_{o \rightarrow d,t,FS}^{median}$, using the estimated coefficients from the regression; Bilateral Transmission is the estimated coefficient on $TransmissionRisk_{o \rightarrow d,t,FS}$, $\hat{\beta}_{o,t}$, with ***, **, and * denoting the statistical significance of $\hat{\beta}_{o,t}$ being different from one; and Regularity of Transmission is the R^2 of the regression. We exclude origin-destination crises that contain fewer than 10 firms from the regressions. Financial Transmission is the ratio of $\hat{\alpha}_{o,t}^{FIN}/\hat{\alpha}_{o,t}$ from a firm-level regression of $CountryRisk_{i,o,t} - CountryRisk_{i,o,t,FS}$ on a constant, $\hat{\alpha}_{o,t}$, and an indicator equal to one if the firm is a financial firm, $\hat{\alpha}_{o,t}^{FIN}$, with ***, **, and * denoting the statistical significance of $\hat{\alpha}_{o,t}^{FIN}$ being different from

(d) Table 7, continued

Normal risk transmission and crisis-specific transmission patterns.

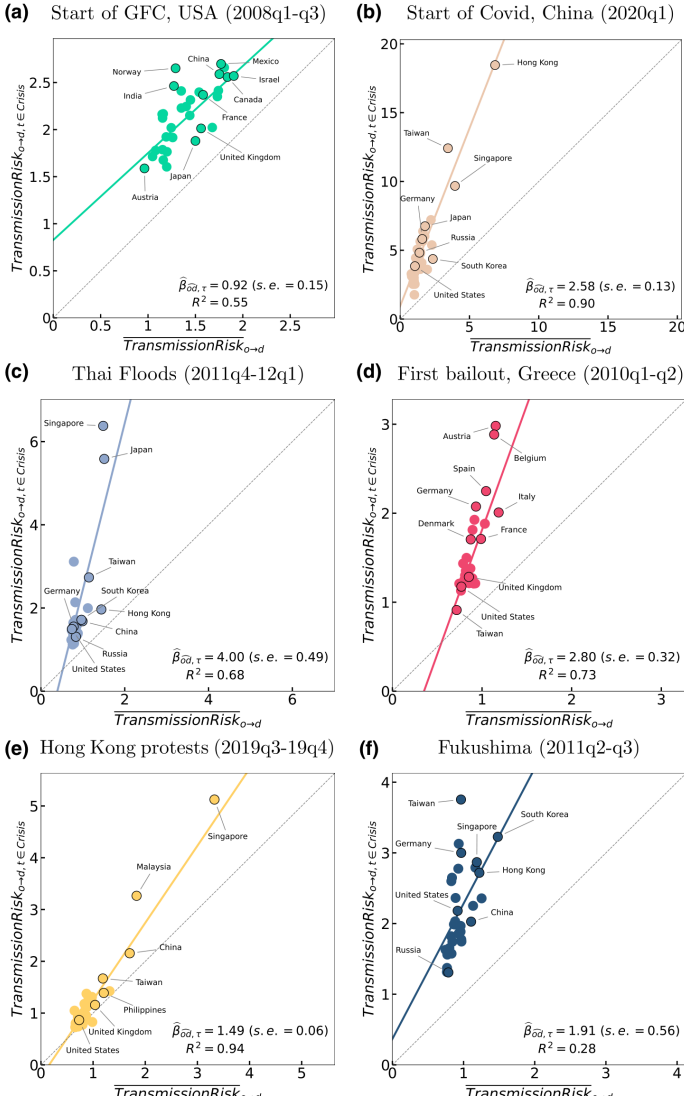
5.7 Figures 7–8: visualizing crisis transmission

Read:

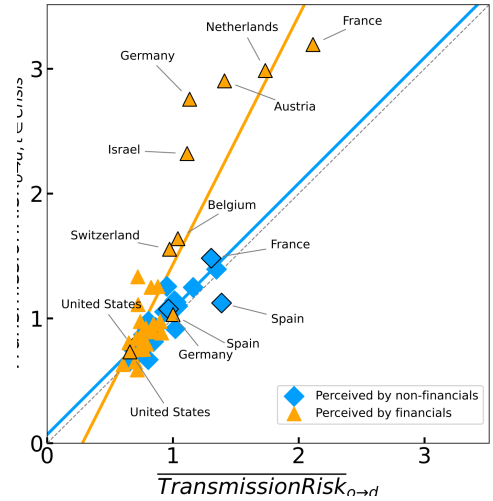
- Figure 7 plots crisis-period transmission against normal-time transmission for six crises.
- The start of COVID in China has very high global impact and high regularity; many countries are pulled upward.
- Thai floods show strong bilateral transmission to countries tied to Thailand and East Asian supply chains.
- Fukushima has low regularity: several countries with little normal exposure react strongly.
- Figure 8 separates financial and non-financial firms for the Italian sovereign debt crisis. Financial firms are far above the 45-degree line, while non-financial firms are not.

Interpretation: Figure 7 makes the meaning of Table 7 intuitive. A crisis near the 45-degree line is a proportional amplification of normal links. A steep slope indicates strong bilateral transmission. A high intercept indicates global impact. Scattered points and low fit indicate irregular contagion. Figure 8 shows why sectoral decomposition matters: Italian sovereign risk is mainly a financial-system transmission event.

Economic message: Crises do not spread through a single universal channel. Some spread through supply chains, some through financial balance sheets, and some through public policy spillovers.



(a) Figure 7



(b) Figure 8

Crisis transmission and financial-sector transmission.

5.8 Tables 8–10: country risk, capital flows, sudden stops, and heterogeneous perceptions

Read:

- Table 8 shows that higher country risk is associated with lower total capital inflows. A one-standard-deviation increase in country risk is associated with a large reduction in inflows relative to the sample mean.
- Crisis dummies produce a similar message: country crises and global crises coincide with lower capital inflows.
- Table 9 shows that country risk is associated with a higher probability of sudden-stop episodes, while global risk is more strongly related to retrenchment.
- Table 10 decomposes perceptions. Foreign firms' perceptions explain inflows more strongly than domestic

perceptions.

- Financial firms' perceptions are especially important for portfolio flows, the most volatile component of capital flows.

Interpretation: Tables 8–10 extend the paper from measurement to macro-financial relevance. Table 8 says country risk affects external financing conditions. Table 9 says risk perceptions are linked to crisis-like capital-flow events. Table 10 clarifies whose beliefs matter: the relevant risk is not simply what domestic firms say about their own country, but what foreign and financial firms perceive about that country.

Economic message: These results blur the push-versus-pull distinction. Country risk is country-specific, but it is often measured by foreign and financial firms. It is therefore a pull factor in location and a push factor in perception.

TABLE 8
Country risk and capital flows

	Total inflows _{it,j} (%)				
PANEL A	(1)	(2)	(3)	(4)	(5)
CountryRisk _{it,j} ^{ALL} (std.)		-0.598*** (0.186)	-0.583*** (0.183)	-0.511*** (0.136)	-0.304* (0.173)
GlobalRisk _{it,j} (std.)	-0.459*** (0.075)	-0.267** (0.099)	-0.279* (0.103)		
Real GDP growth _{it,j}			-0.003 (0.007)	0.025*** (0.009)	
CountrySentiment _{it,j} ^{ALL} (std.)					0.652*** (0.191)
R ²	0.122	0.132	0.137	0.275	0.269
N	2,936	2,936	2,796	2,796	2,936
PANEL B	Total inflows _{it,j} (%)				
	(1)	(2)	(3)	(4)	
1(CountryCrisis _{it,j})		-1.499*** (0.362)	-1.381*** (0.340)	-0.888*** (0.309)	
1(GlobalCrisis _{it,j})	-1.671*** (0.241)	-1.705*** (0.243)	-1.830*** (0.237)		
Real GDP growth _{it,j}			-0.003 (0.007)	0.026*** (0.009)	
R ²	0.100	0.111	0.118	0.271	
N	2,936	2,936	2,796	2,796	
Country FE	yes	yes	yes	yes	yes
Time FE	no	no	no	yes	yes

Notes: This table shows coefficient estimates and standard errors from regressions at the country-quarter level. All other variables are defined as shown in Table 4. Standard errors are clustered at the country level. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(a) Table 8

TABLE 9
Country risk, sudden stops, and retrenchment

	1(Stop episode for total flows _{it,j})				
PANEL A	(1)	(2)	(3)	(4)	(5)
CountryRisk _{it,j} ^{ALL} (std.)		0.082** (0.037)	0.079** (0.038)	0.086** (0.034)	0.060* (0.032)
GlobalRisk _{it,j} (std.)	0.092*** (0.009)	0.066*** (0.015)	0.067*** (0.015)		
Real GDP growth _{it,j}			-0.001 (0.001)	0.000 (0.001)	
CountrySentiment _{it,j} ^{ALL} (std.)					-0.070** (0.028)
R ²	0.088	0.095	0.096	0.337	0.342
N	2,734	2,734	2,627	2,627	2,734
PANEL B	1(Retrenchment episode for total flows _{it,j})				
	(1)	(2)	(3)	(4)	(5)
CountryRisk _{it,j} ^{ALL} (std.)		0.009 (0.025)	0.007 (0.024)	0.012 (0.020)	-0.006 (0.020)
GlobalRisk _{it,j} (std.)	0.007*** (0.010)	0.064** (0.013)	0.064** (0.014)		
Real GDP growth _{it,j}			-0.001 (0.001)	0.000 (0.001)	
CountrySentiment _{it,j} ^{ALL} (std.)					-0.045 (0.029)
R ²	0.053	0.053	0.053	0.264	0.262
N	2,734	2,734	2,627	2,627	2,734
Country FE	yes	yes	yes	yes	yes
Time FE	no	no	no	yes	yes

Notes: This table shows coefficient estimates and standard errors from regressions at the country-quarter level. The outcome in Panel A, 1(Stop episode for total flows_{it,j}), is a dummy equal to one if there is a stop episode for total capital flows of country *i* in quarter *t*. The outcome in Panel B, 1(Retrenchment episode for total flows_{it,j}), is a dummy equal to one if there is a retrenchment episode for total capital flows of country *i* in quarter *t*. ***, **, and * denote significance at the 1%, 5%, and 10% level, respectively.

(b) Table 9

TABLE 10
Capital flows and heterogeneous perceptions of country risk

	Total inflows _{it,j} (%)			
PANEL A	(1)	(2)	(3)	(4)
CountryRisk _{it,j} ^{ALL} (std.)	-0.551*** (0.145)			
CountryRisk _{it,j} ^{US firms} (std.)		-0.781*** (0.183)		
CountryRisk _{it,j} ^{NH} (std.)			-0.446*** (0.160)	-0.423** (0.162)
WFI _{it,j} (std.)				-0.091* (0.046)
R ²	0.260	0.259	0.258	0.260
N	2,936	2,936	2,936	2,936
PANEL B	Total inflows _{it,j} (%)		Portfolio _{it,j} (%)	
	(1)	(2)	(3)	(4)
CountryRisk _{it,j} ^{NH} (std.)	-0.430*** (0.159)	-0.418*** (0.144)		
CountryRisk _{it,j} ^{NH} (std.)			0.065 (0.065)	
FirmRisk _{it,j} ^{ALL} (std.)		-0.149** (0.067)		
CountryRisk _{it,j} ^{FIN} (std.)			-0.332*** (0.116)	-1.100** (0.424)
CountryRisk _{it,j} ^{FIN} (std.)				-0.261 (0.160)
R ²	0.287	0.361	0.261	0.134
N	2,710	2,163	2,936	2,936
Country FE	yes	yes	yes	yes
Time FE	yes	yes	yes	yes

(c) Table 10

Country risk and international capital allocation.

6 Short summary

- The paper constructs a firm-country-quarter measure of perceived country risk from global earnings calls.
- It uses country-specific training libraries to detect country-related language and counts this language near risk and uncertainty words.
- Country exposure, sentiment, country risk, transmission risk, and global risk are distinct measures.
- The exposure measure validates against headquarters, exports, and subsidiaries.
- The risk measure validates against equity returns, CDS spreads, and realized volatility.
- The paper identifies two global crises in the sample: the GFC and COVID.
- It identifies many local crises, including sovereign-debt crises, natural disasters, protests, coups, trade disputes, oil strikes, and financial shocks.
- Normal risk transmission follows a gravity pattern.
- Crisis transmission can be global, bilateral, irregular, financial, or non-financial.
- Developed-market and sovereign-debt crises often transmit through financial firms.
- Emerging-market crises and natural disasters often transmit through exposed non-financial firms and supply chains.

- Higher perceived country risk is associated with lower capital inflows.
- Higher perceived country risk is associated with greater sudden-stop risk.
- Foreign firms’ and financial firms’ perceptions are particularly informative for capital flows.

7 Conclusion

7.1 Core conclusion

Hassan, Schreger, Schwedeler, and Tahoun show that country risk can be measured at a granular level by reading what managers and investors say about countries in earnings calls. This creates a bridge from micro-level risk perceptions to macro-level outcomes: asset prices, capital flows, sudden stops, and international contagion.

7.2 Conceptual contribution

The paper’s main conceptual contribution is to treat country risk as a perceived and transmitted object rather than a latent aggregate shock. A country is risky not only because asset prices fall or spreads rise, but because identifiable agents around the world discuss risks associated with that country. This distinction matters because the same country-risk event can look different depending on who perceives it.

7.3 Measurement contribution

The paper’s micro-to-macro design is powerful for three reasons. First, it separates exposure, risk, and sentiment. Second, it allows flexible aggregation by firm type and location. Third, it can identify sources of risk by returning to the underlying text snippets. This makes the data useful both for econometrics and for narrative event analysis.

7.4 Interpretation of the main findings

The central empirical message is that country-risk transmission is usually structured but sometimes unstable. In normal times, risk transmission looks like gravity: proximity, trade, language, and historical ties matter. In crises, the map can shift. A global financial shock or pandemic can make distant countries care about the same origin. A natural disaster can transmit through supply chains. A sovereign-debt crisis can transmit through bank balance sheets. A political event can transmit through policy expectations even when direct trade links are weak.

7.5 Implications for international finance

The results enrich the push–pull framework of capital flows. Country risk is a pull factor because it is tied to a destination country. Yet the most informative perceptions often come from foreign and financial firms, which makes it resemble a push factor in the sense that outside investors and global intermediaries decide how risky a destination looks. The paper therefore suggests that capital-flow instability can arise from interactions between local events and global perception networks.

7.6 Implications for policymakers

For policymakers, the measures provide an early diagnostic tool. A country can monitor whether its perceived risk is rising among domestic firms, foreign firms, or financial firms. These distinctions matter: foreign-firm concern may predict capital withdrawal, while financial-firm concern may signal portfolio-flow stress. The measures also help distinguish local crises from global crises and real-side transmission from financial contagion.

7.7 Implications for firms and investors

For firms, the paper suggests that risk management should track not only direct sales exposure but also narrative exposure: how crises in other countries are discussed by peers, suppliers, customers, and financial intermediaries. For investors, the text measures provide a way to detect whether a country-risk event is becoming systemic, bilateral, or sector-specific.

7.8 Limitations and open questions

The measures capture the perceptions of listed firms that hold English-language calls, so they are less informative about private firms, households, and purely domestic sectors. They also capture perceived risk, not necessarily objective risk. A useful future direction is to compare text-based perceptions with realized macro shocks, policy responses, and private-firm outcomes. Another direction is to use the microdata to build predictive models of crises, capital-flow reversals, or sectoral spillovers.

7.9 Takeaway

The paper’s enduring takeaway is that country risk is observable in language before it is fully summarized in macro aggregates. By measuring who is worried, about which country, and through which channel, the paper turns international risk transmission into a measurable network of beliefs and exposures.